

Loan Approval Prediction

Initial Draft

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DSA5103 Intelligent Data Analytics

Date: November 26, 2024

Introduction to the problem

Loan approval is a critical process for financial institutions, directly impacting their ability to manage risk and allocate financial resources efficiently. Evaluating loan applications involves assessing the credit risk associated with applicants based on several factors, including their financial standing, credit history, and demographic information. Making accurate loan approval decisions not only safeguards the financial stability of the lender but also ensures fair access to credit for eligible applicants.

The **Loan Approval Project** aims to analyze a comprehensive dataset containing detailed credit risk and loan application information. By developing a machine learning model, this project seeks to classify loan applications as "approved" or "not approved." The dataset includes metadata and critical attributes about applicants that influence loan approval decisions. Through exploratory data analysis (EDA) and advanced predictive modeling, this project will provide insights into key drivers of loan approval and optimize decision-making processes.

Data Description

The dataset has 11 predictors and 1 target variables. Among the 11 predictors, 7 are numeric variables and 4 are categorical variables.

Key Highlights of the Dataset:

- Applicant Demographics:
 - Variables such as `person_age` and `person_gender` provide context to the profile of loan applicants.
- Financial Attributes:
 - Features like `person_income` (annual income), `person_home_ownership`, `loan_amnt` (loan amount requested), `loan_int_rate` (interest rate), and `person_emp_length` (employment length in years) help gauge the applicant's financial health.
- Credit History:
 - Data on `cb_person_cred_hist_length` (credit history length), `cb_person_default_on_file` (whether the applicant has a history of defaults), and other repayment-related variables serve as indicators of an applicant's creditworthiness.
- Loan Details:
 - Information such as `loan_intent` (purpose of the loan), `loan_amnt` (requested loan amount), and other relevant details add depth to the decision-making process.
- Loan Status:
 - The target variable, `loan_status`, indicates whether the loan application was approved (typically encoded as "1") or rejected (typically encoded as "0").

Data preprocessing updates

For logistic regression and DBSCAN clustering, no new updates were done to the data pre-processing tasks like handling missing values and outliers. The four models GLM, LASSO, GBM and Random Forest had an updated data preprocessing. Only the updated version is discussed in this report.

Missing Values

Only two variables `person_emp_length` (2.75%) and `loan_int_rate` (9.56%) have missing values. The missing values were imputed using the predictive mean matching (PMM) instead of kNN mentioned in initial data analysis report.

For `person_emp_length`, relevant variables include `person_age`, `person_income`, and `cb_person_cred_hist_length`.

For `loan_int_rate`, relevant variables include `loan_amnt`, `person_income`, `loan_grade`, and `cb_person_cred_hist_length`.

Outliers

Cases where a person's employment length in years is much much bigger than their actual age were removed.

Additionally,

- Remove records for borrowers under 70 years old. In practical business scenarios, 70 years is typically the maximum age limit for most bank loans.
- Ensure the borrower's age minus their employment length is at least 18 years. If not, it implies the borrower started working before reaching adulthood, which is illegal in most countries.

Feature transformation

Comprehensive Data Preparation Plan

- Merge Low-Frequency Categories
 - Combine `person_home_ownership` values OWN and OTHER into a single category OTHER due to low frequencies.
 - Combine `loan_grade` values F and G into a single category OTHER for the same reason.
- Convert Categorical Variables to Numerical Format
 - Convert `person_home_ownership` and `loan_intent` into numerical format using one-hot encoding or label encoding as appropriate.
- Transform `loan_grade`
 - Convert `loan_grade` into an ordinal factor reflecting the grade hierarchy (e.g., A < B < C < D < E < OTHER).
 - Convert the ordinal factor into a numerical variable.
- Transform `cb_person_default_on_file`

- Convert `cb_person_default_on_file` into an ordinal factor (No = 0, Yes = 1).
 - Convert the ordinal factor into a numerical variable.
5. Remove Redundant Columns
- Remove the original categorical columns: `person_home_ownership` , `loan_intent` , `loan_grade` , and `cb_person_default_on_file` , as their numerical representations have been created.
6. Apply Log Transformation to Numerical Variables
- Apply logarithmic transformation to variables with skewed distributions to approximate normality: `person_age` , `person_income` , `person_emp_length` , `loan_amnt` , `loan_int_rate` , `cb_person_cred_hist_length` (rename as `log_cred_hist_length` and remove the original variable).
7. Add a New Feature
- Create `loan_percent_income` as the ratio of `loan_amnt` to `person_income`.
8. Normalize and Standardize Numerical Variables
- Normalize all numerical variables to a 0-1 range for comparability.
 - Standardize (z-score normalization) to have a mean of 0 and a standard deviation of 1 for use in distance-based models.
9. Remove Zero-Variance Variables
- Identify and remove any variables with zero variance, as they provide no information for modeling.
10. Filter Variables Based on Correlation
- Select variables with an absolute correlation coefficient greater than 0.2 to the target variable.
11. Explore Interaction Terms Create and evaluate interaction terms for potential improvements in predictive performance:
- `loan_percent_income * log_income`
 - `log_int_rate * loan_grade_numeric`
 - `person_home_ownership_RENT * loan_percent_income`

Description of Modelling Approach

Supervised Learning

Four different models were used (GLM, LASSO, GBM, Random Forest) to predict the target variable. Each model is trained using three different sets of independent variables:

1. All Variables:
 - Includes all original variables after data preparation, without adding any interaction terms.
2. Selected Variables + Interaction:
 - Includes variables selected based on the heatmap analysis and their interaction terms.
3. All Variables + Interaction:
 - Includes all original variables after data preparation, along with additional interaction terms.

GLM (Generalized Linear Model)

Firstly, the GLM was chosen as the baseline model. Logistic regression is a widely used statistical model, particularly suited for classification tasks. The coefficients output by the model directly explain the relationship between variables and the target, which is highly beneficial for understanding business insights. With its low computational complexity, it is faster than more complex models like Random Forest or Gradient Boosting, making it ideal for handling small to medium-sized datasets (as this dataset qualifies). As a baseline model, the GLM often provides a reasonable performance benchmark, allowing for further optimization or replacement with more complex models if needed.

LASSO (Least Absolute Shrinkage and Selection Operator)

Secondly, Lasso regression is a linear model that adds an L1 regularization term to the loss function, enabling feature selection and preventing overfitting. The L1 regularization in Lasso regression shrinks some variable coefficients to zero, automatically performing feature selection and removing irrelevant features, thereby simplifying the model. Regularization not only limits model complexity but also enhances generalization ability. This is especially effective in situations with multicollinearity or limited samples, where it prevents overfitting. Since unimportant features are entirely excluded, the Lasso model becomes easier to interpret, making it clear which variables significantly influence the target variable. The strength of regularization is controlled by the hyperparameter λ (lambda), which can be optimized via cross-validation to balance model complexity and predictive performance.

GBM (Gradient Boosting Machine)

Then, GBM offers powerful nonlinear modeling capabilities and can capture complex nonlinear relationships effectively. GBM is highly adaptable, supporting various tasks such as classification and regression. By tuning hyperparameters like `n.trees`, `interaction.depth`, and `shrinkage`, GBM can be adjusted to meet the needs of different datasets. As an iterative optimization-based model, GBM reduces residuals at each step, continually improving performance, and typically achieving higher prediction accuracy than standalone models like GLM and Lasso. Additionally, GBM can evaluate the importance of each variable in the model's predictions, facilitating result analysis and interpretation.

Random Forest

Finally, Random Forest, an ensemble of multiple decision trees, provides high prediction accuracy, particularly for handling complex and high-dimensional data. Random Forest is robust against outliers and noise, as its results are based on the average predictions (for regression) or voting (for classification) across multiple trees. It does not require assuming a linear relationship between variables and the target, making it capable of capturing nonlinear relationships effectively. By calculating feature importance (e.g., based on Gini impurity or split criteria), it can automatically evaluate the significance of variables, supporting interpretability and feature selection. Furthermore, it is less sensitive to correlations among features, allowing it to perform stably even in the presence of multicollinearity.

Model Parameter Settings

GLM		
1	10-fold cross-validation was configured using trainControl.	method = "cv", number = 10
2	Class probability output was enabled.	classProbs = TRUE
3	A custom evaluation function was implemented.	summaryFunction = custom_summary
4	GLM model based on logistic regression was constructed.	family = binomial(link = "logit")
LASSO		
1	10-fold cross-validation was configured using trainControl.	The same as above
2	Class probability output was enabled.	The same as above
3	A custom evaluation function was implemented.	The same as above
4	A tuneGrid was set to define the search range for the penalty parameter lambda.	The range of lambda was set from 0.001 to 0.1, with a step size of 0.01.
5	The model optimizes based on different lambda values and automatically selects the optimal parameter.	The lambda with the highest ROC value The lambda with the highest accuracy
GBM		
1	10-fold cross-validation was configured using trainControl.	The same as above
2	Each validation randomly splits the data to ensure model performance stability.	
3	Class probability output was enabled.	The same as above
4	A custom evaluation function was implemented.	The same as above
RANGER FOREST		
1	Repeated cross-validation	method = 'repeatedcv', number = 10, repeats = 5
2	Randomly split the data into 10 folds, selecting different training and testing sets for each round.	
3	Class probability output was enabled.	The same as above
4	A custom evaluation function was implemented.	The same as above
5	Perform a random search within the parameter space to find the optimal parameter combination.	search = 'random'
6	Set the number of trees in the forest to 200 to enhance model stability.	num.trees = 200

Logistic Regression

It a supervised learning technique used mainly for binary classification. It predicts the probability of an outcome event and classifies them by putting a threshold on the probability values. If the probability of outcome is greater than 0.5, it classifies the item into say class 1 and into class 0 otherwise. It uses logistic function to predict the probabilities. This uses logarithm of the odds to predict the relationship between the input variables and binary output variables. In order to test the modelling efficiency, the dataset was divided into training and testing set such that 80% of the dataset was used was testing and the remaining 20% for testing. Only original variables without adding any interaction terms was used. The data preparation techniques mentioned in initial data analysis were used for this modelling.

Unsupervised Learning

Density Based Spatial Clustering of Applications with Noise (DBSCAN)

It is an unsupervised clustering algorithmn that groups data points based on their density in space. This is particularly useful when the clusters are of varying shapes and sizes unlike in k-means clustering where all clusters are assumed to be hyperspherical in shape and about equal sizes. Only numeric variables with data preparation mentioned in initial data analysis was used. Here the entire dataset is used to study how well the clusters can predict loan approval data points from those that weren't approved.

The parameters required for DBSCAN to work are `minPts` and `eps` . A point is a core point if there are atleast `minPts` within a radius of `eps` around it. Any point that is within `eps` of a core point is density-reachable. Clusters are formed by grouping all points that are density-reachable from at least one core point. Any point that is not density reachable from any core point is labelled either as noise or outlier.

A general rule of thumb is `minPts = dimensions + 1` , here, dimensions refer to the number of features in your dataset. Small `eps` results in many small clusters and large `eps` results in a few large clusters. To determine the value of `eps` , a k-distance plot was used. It computes distances to the k-th nearest neighbor for all points where `k = minPts - 1` . `minPts = 9` in this case as 8 input numeric features were used, the value of `k` is then set to 8. These distances are plotted in ascending order and the elbow point of this plot is used to estimate the value for `eps` .

Initial Results

Following statistics are studied to measure each model's performance.

- ROC** measures the overall classification ability of the model. The closer it is to 1, the better the model performs.
- Sensitivity** measures the model's ability to correctly detect positive cases. Higher values indicate better performance.
- Specificity** measures the model's ability to correctly identify negative cases. Higher values indicate better performance.
- Accuracy** reflects the overall classification performance of the model. Higher values indicate better accuracy.
- Kappa** (Classification Consistency) measures the reliability of the classification results. Higher values indicate better consistency.

Supervised learning - Logistic Regression

The data pre-processing used for this modelling is the same as mentioned in the initial data analysis report. No new data preprocessing was used to obtain the results below.

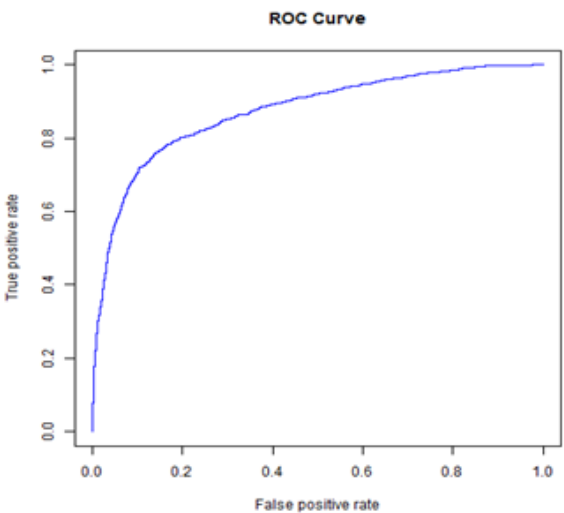
- Accuracy:** 86.5%, meaning it correctly predicted the loan status 86.5% of the time. By looking at the **No Information Rate (NIR)** of 78.05% which suggests that model can be improved by taking into account the interactions.
- True Positive Rate (TPR):** 46.71%. The model was able to correctly identify 46.71% of the time, that is very low which suggests the model struggles with the minority classes i.e. applications with loan approved.
- True Negative Rate (TNR):** 97.21%. that means the model correctly identifies 97.21% of the negatives. This suggest model is biased in predicting negatives correctly

but poor on predicting the positive. This implies there is a need for improvement in the model.

Balanced Accuracy: 76.0%, balancing TPR and TNR.

The model’s **kappa** value was found to be 0.505. The value is low indicating a room for the improvement in the model. It may be improved by optimizing the model, tuning the hyperparameters and adding interaction terms.

ROC curve which is a plot between true positive rate and true negative rate is plotted. The area under the curve was found to be 0.874, which indicates a good ability to for loan approval prediction.

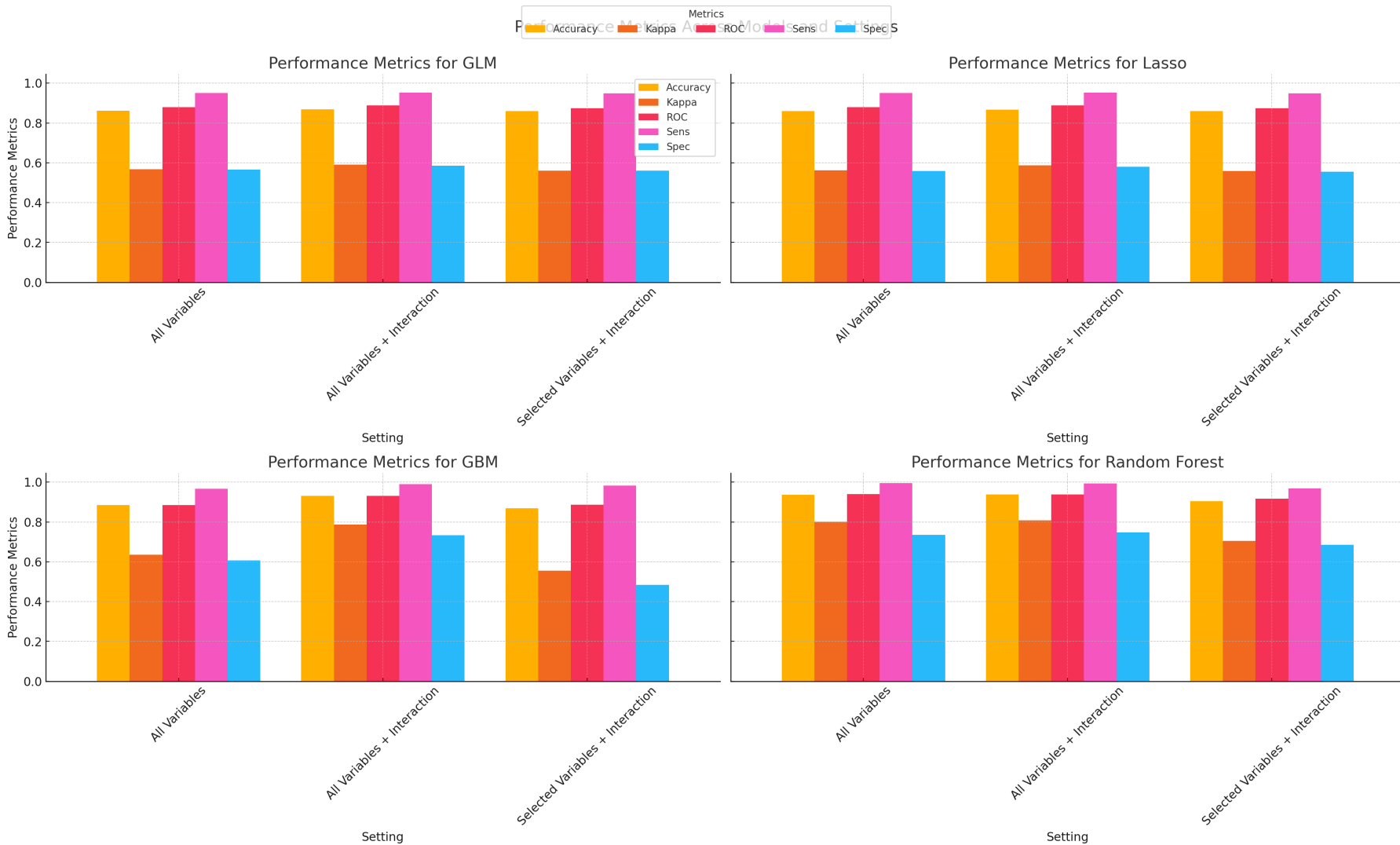


Supervised learning - GLM, LASSO, GMB and Random Forest

Let's look at the combined results table for all the four models and then discuss each statistic one by one.

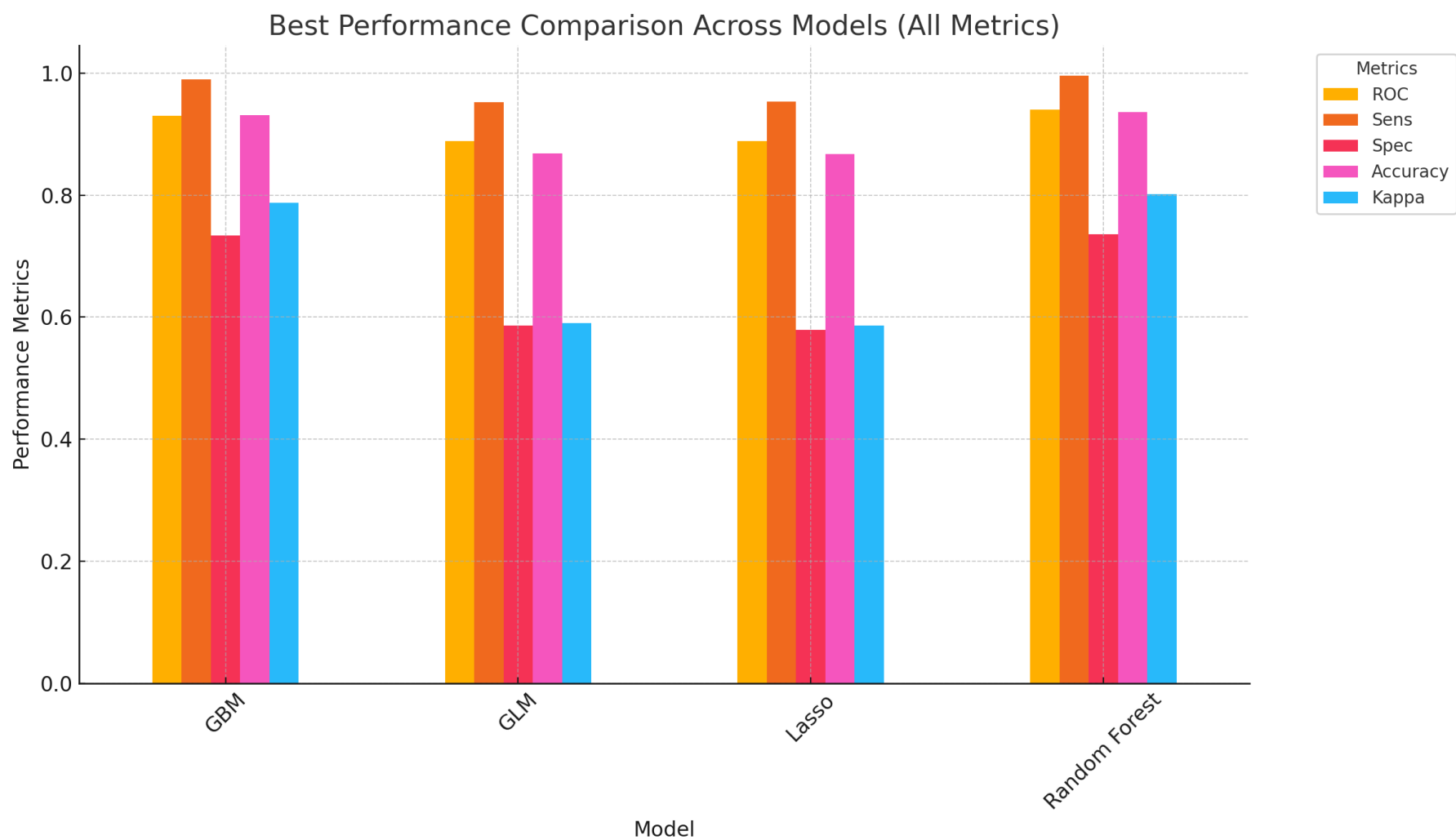
Method	Setting	ROC	Sens	Spec	Accuracy	Kappa	ROCSD	SensSD	SpecSD	AccuracySD	KappaSD
GLM	All Variables	0.879036	0.949706	0.565142	0.861636	0.56777	0.010154	0.005215	0.021762	0.006909	0.022928
GLM	Selected Variables + Interaction	0.873912	0.948395	0.560385	0.859535	0.561097	0.005204	0.00484	0.022468	0.004122	0.016047
GLM	All Variables + Interaction	0.88806	0.952276	0.585979	0.868389	0.590788	0.004814	0.004833	0.019935	0.004486	0.015639
Random Forest	All Variables	0.940566	0.995763	0.735629	0.936188	0.801776	0.007234	0.001266	0.019334	0.004718	0.016059
Random Forest	Selected Variables + Interaction	0.917576	0.967999	0.686089	0.903437	0.705093	0.008241	0.003254	0.020279	0.005395	0.018049
Random Forest	All Variables + Interaction	0.937847	0.994179	0.747173	0.937612	0.807494	0.003962	0.001492	0.015262	0.003444	0.011631
Lasso	All Variables	0.879082	0.949916	0.558611	0.860303	0.562339	0.00634	0.005094	0.021762	0.004498	0.016647
Lasso	Selected Variables + Interaction	0.874039	0.949653	0.55597	0.859494	0.559508	0.005195	0.005378	0.021879	0.0038	0.014677
Lasso	All Variables + Interaction	0.888305	0.953115	0.579447	0.86754	0.586638	0.004745	0.004872	0.021418	0.004837	0.017174
GBM	All Variables	0.885536	0.966697	0.607158	0.884361	0.636151	0.005787	0.006422	0.042849	0.007304	0.030048
GBM	Selected Variables + Interaction	0.885933	0.982903	0.483585	0.868552	0.555451	0.006359	0.00738	0.045786	0.006311	0.031812
GBM	All Variables + Interaction	0.930197	0.989773	0.733581	0.931102	0.787502	0.004959	0.002526	0.017821	0.003718	0.012712

The performance metrics of each model under different settings are compared as follows:



In summary, the performance metrics of GBM and RangerForest are significantly better than those of LASSO and GLM. GLM has the lowest performance, while GBM and RangerForest show highly similar results with no significant differences.

A horizontal comparison of the performance metrics under the optimal settings of each model further validates the above conclusions.

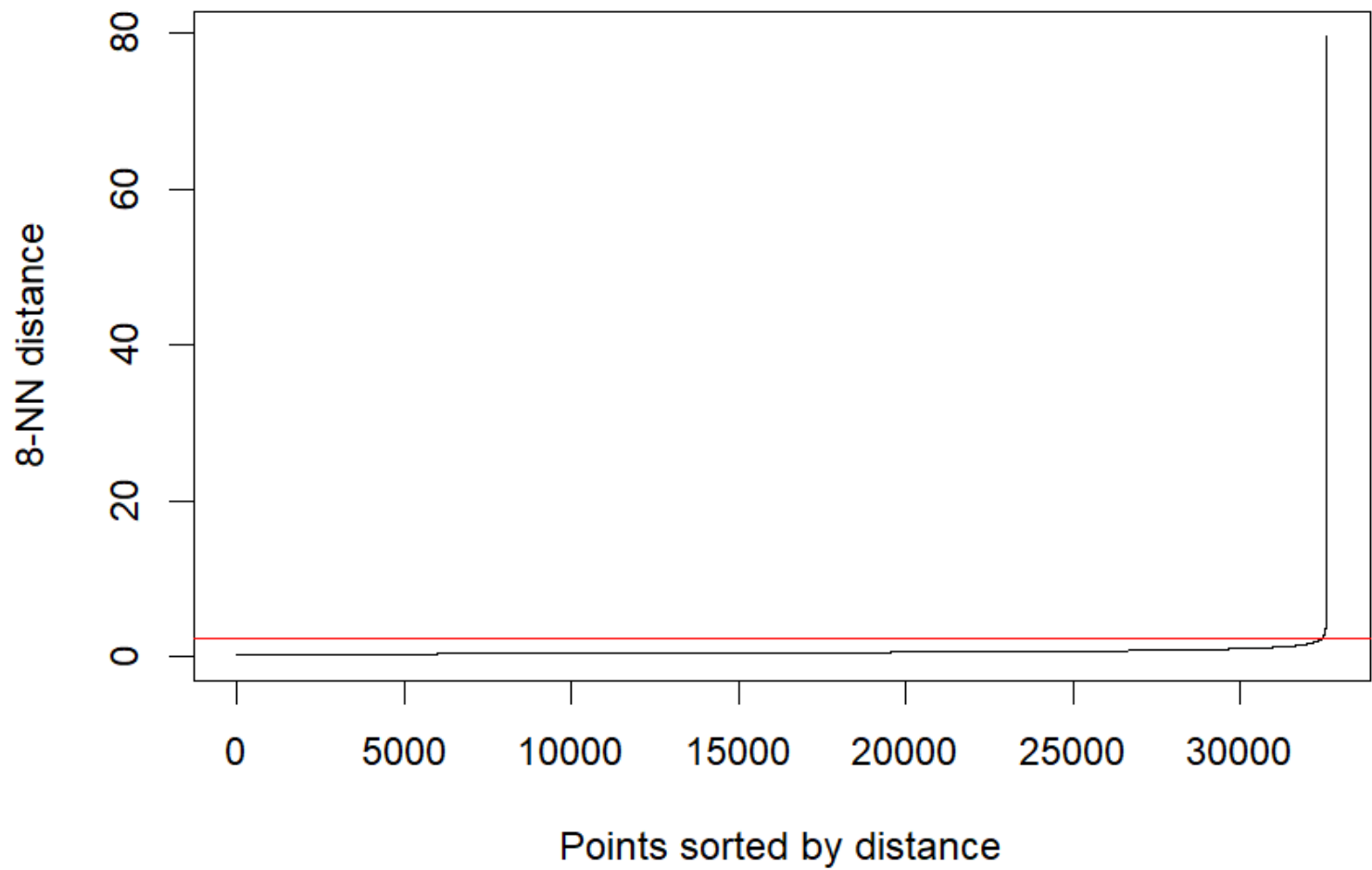


Random Forest (All Variables or All Variables + Interaction) performs exceptionally well across most metrics and is the most recommended model. GBM (All Variables + Interaction) performs excellently in accuracy and classification consistency, making it a strong alternative. GLM needs further improvement, particularly in negative case detection (Specificity). Adding interaction terms significantly enhances the performance of most models, such as GBM and Random Forest.

Unsupervised learning - Density Based Spatial Clustering of Applications with Noise (DBSCAN)

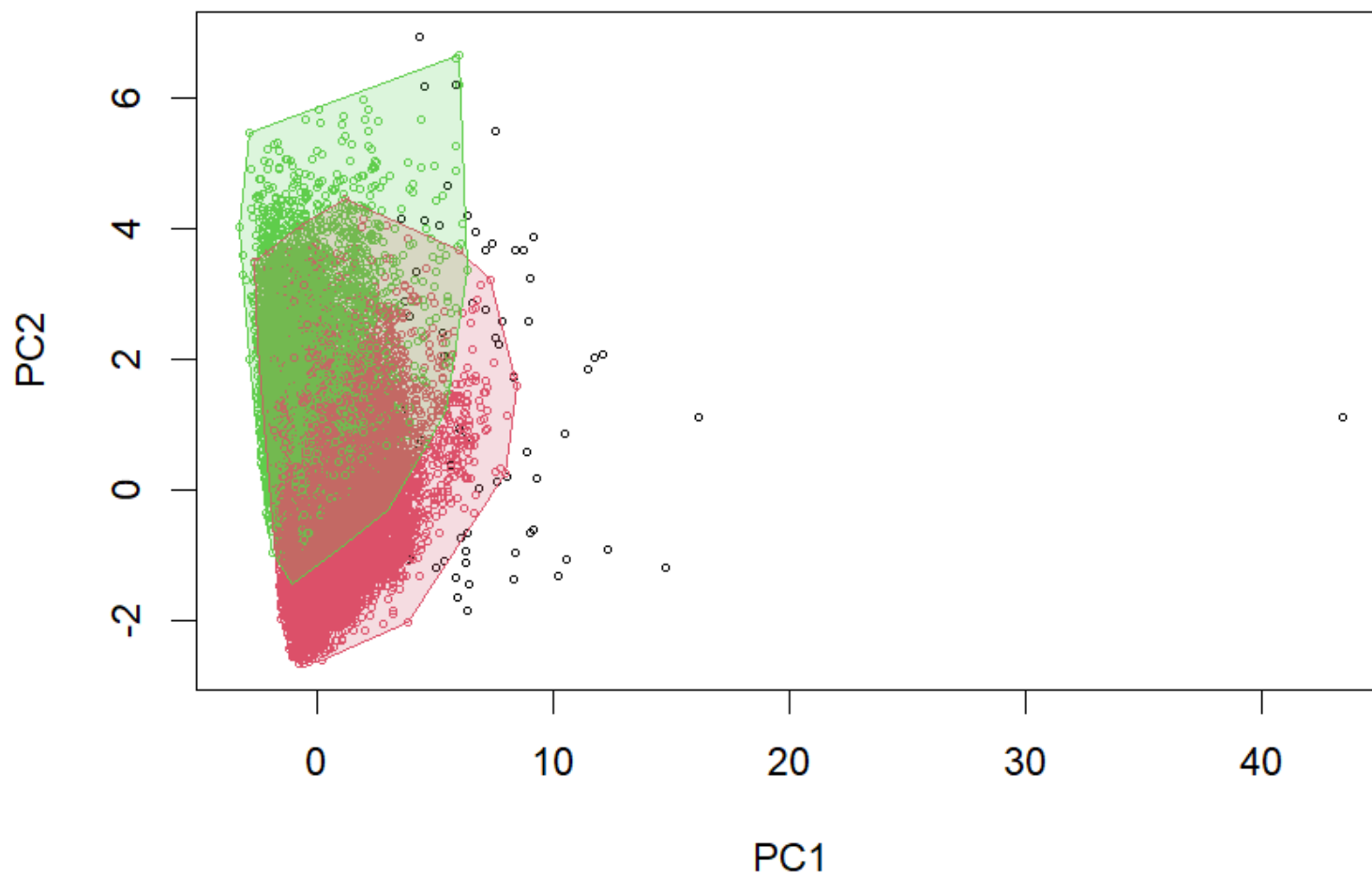
The data pre-processing used for this modelling is the same as mentioned in the initial data analysis report. No new data preprocessing was used.

Using the 8-NN distance plot below, the value of `eps` was estimated to be 2.3, which was then used to do the clustering using `dbscan()` function.



The clusters are visualized using principal components that show the irregular shapes and sizes of the two clusters. Look at the Hullplot below

Convex Cluster Hulls



Using euclidean distances, $\text{eps}=2.3$ and $\text{minPts}=9$, the DBSCAN clustering estimated existence of two imbalanced clusters and 71 noise points/outliers. Cluster 1 has 25424 points and cluster 2 has 7084 points.

DBSCAN clustering for 32579 objects.
Parameters: $\text{eps} = 2.3$, $\text{minPts} = 9$
Using euclidean distances and $\text{borderpoints} = \text{TRUE}$
The clustering contains 2 cluster(s) and 71 noise points.

```
0      1      2
71 25424 7084
```

These predictions are further compared with the existing target variable `loan_status` using the table below. The row value 0 corresponds to points where the loan was not approved and 1 corresponds to those applications where the loan was approved. DBSCAN clustering identifies these clusters such that it predicts the loan approval correctly ~99.68% of the times and it predicts the loan rejection correctly ~99.81% of the times.

```
0      1      2
0      48 25424      0
1      23      0 7084
```

Let's look at top datapoints belonging to the two clusters and noise points.

10 data points corresponding to cluster 1 with 25424 total data points

	person_age	person_income	person_emp_length	loan_amnt	loan_int_rate	loan_status	loan_percent_income	cred_hist_length	dbCluster
2	21	9600	5	1000	11.14	0	0.10	2	1
15	23	115000	2	35000	7.90	0	0.30	4	1
16	23	500000	7	30000	10.65	0	0.06	3	1
17	23	120000	0	35000	7.90	0	0.29	4	1
21	25	162500	2	35000	7.49	0	0.22	4	1
22	25	137000	9	34800	16.77	0	0.25	2	1
24	24	10980	0	1500	7.29	0	0.14	3	1
28	23	11000	0	1400	9.32	0	0.13	3	1
32	25	120000	2	32000	6.62	0	0.27	2	1
34	25	306000	2	24250	13.85	0	0.08	3	1

10 data points corresponding to cluster 2 with 7084 total data points

	person_age	person_income	person_emp_length	loan_amnt	loan_int_rate	loan_status	loan_percent_income	cred_hist_length	dbCluster
3	25	9600	1	5500	12.87	1	0.57	3	2
4	23	65500	4	35000	15.23	1	0.53	2	2
5	24	54400	8	35000	14.27	1	0.55	4	2
6	21	9900	2	2500	7.14	1	0.25	2	2
7	26	77100	8	35000	12.42	1	0.45	3	2
8	24	78956	5	35000	11.11	1	0.44	4	2
9	24	83000	8	35000	8.90	1	0.42	2	2
10	21	10000	6	1600	14.74	1	0.16	3	2
11	22	85000	6	35000	10.37	1	0.41	4	2
12	21	10000	2	4500	8.63	1	0.45	2	2

10 data points corresponding to noise/outliers

	person_age	person_income	person_emp_length	loan_amnt	loan_int_rate	loan_status	loan_percent_income	cred_hist_length	dbCluster
82	144	250000	4	4800	13.57	0	0.02	3	0

184	144	200000	4	6000	11.86	0	0.03	2	0
576	123	80004	2	20400	10.25	0	0.25	3	0
748	123	78000	7	20000	11.11	0	0.26	4	0
17834	32	1200000	1	12000	7.51	0	0.01	8	0
17835	34	948000	18	2000	9.99	0	0.00	7	0
17841	35	648000	6	10000	17.19	1	0.02	8	0
17848	35	500000	17	20000	13.85	0	0.04	7	0
17850	33	468000	14	3750	15.62	1	0.01	10	0
25715	32	703800	14	12000	15.33	1	0.01	9	0

In **cluster 1**, most applicants are young and in their 20's, the income is either very high or very low, the high income also corresponds to zero employment length in most cases implying that the person must own their own business, however, the most important feature is percentage of money allocated towards loan repayment which is low considering the high income in most cases.

In **cluster 2**, applicants are mostly young and in their 20's, with an moderate amount of income in most cases, relatively higher length of employment on average compared to cluster 1. However, the percentage of money allocated towards loan repayment is quite high compared to cluster 1.

For **noise points**, the applicants are either very old or of moderate age, in their 30's, with exceptionally high incomes and very low percent of money allocated towards loan repayment. Thus, these applications need further investigations since these are anamolies.

This algorithms works exceptionally well in understanding the underlying patterns especially for imbalanced clusters with varying shapes and sizes. Given a new application, it should be able to predict the outcome of loan approval process accurately in most of the cases except when the applications resemble the patterns derived in the anamolies above.

Contribution page

- Mayuri led the project proposal and finalization, data preprocessing and initial exploratory analysis and unsupervised modelling. She has and will lead the documentation/report writing part and will write conclusions/interpretations for the models.
- Seunghyun primarily focused on data preprocessing tasks, including handling missing values and exploring imputation strategies. Additionally, he contributed towards drafting the introduction, providing a detailed data description, and conducting exploratory data analysis (EDA) for the report.
- Gen led the supervised learning component by using four different models and hyperparameter optimization, actively worked on data preprocessing, exploratory data analysis and contributed to the corresponding document writing.
- Manisha worked on data preprocessing by handling missing values, outliers and worked on data visualization. She also worked on creating data quality report, developing the logistic regression model and helped in interpreting the results from the model.

Appendix: Data Quality Report

Numerical Variables

variable	n	missing	missing_pct	unique	unique_pct	mean	min	Q1	median	Q3	max	sd
person_age	32581	0	0.00	58	0.178	27.735	20.00	23.00	26.00	30.00	144.00	6.348
person_income	32581	0	0.00	4295	13.183	66074.848	4000.00	38500.00	55000.00	79200.00	6000000.00	61983.119
person_emp_length	32581	895	2.75	37	0.114	4.790	0.00	2.00	4.00	7.00	123.00	4.143
loan_amnt	32581	0	0.00	753	2.311	9589.371	500.00	5000.00	8000.00	12200.00	35000.00	6322.087
loan_int_rate	32581	3116	9.56	349	1.071	11.012	5.42	7.90	10.99	13.47	23.22	3.240
loan_status	32581	0	0.00	2	0.006	0.218	0.00	0.00	0.00	0.00	1.00	0.413
loan_percent_income	32581	0	0.00	77	0.236	0.170	0.00	0.09	0.15	0.23	0.83	0.107
cb_person_cred_hist_length	32581	0	0.00	29	0.089	5.804	2.00	3.00	4.00	8.00	30.00	4.055

Categorical Variables

variable	n	missing	missing_pct	unique	unique_pct	freqRatio	1st mode	1st mode freq	2nd mode	2nd mode freq	least common	least common freq
person_home_ownership	32581	0	0	4	0.012	1.22	RENT	16446	MORTGAGE	13444	OTHER	107
loan_intent	32581	0	0	6	0.018	1.06	EDUCATION	6453	MEDICAL	6071	HOMEIMPROVEMENT	3605
loan_grade	32581	0	0	7	0.021	1.03	A	10777	B	10451	G	64
cb_person_default_on_file	32581	0	0	2	0.006	4.67	N	26836	Y	5745	Y	5745